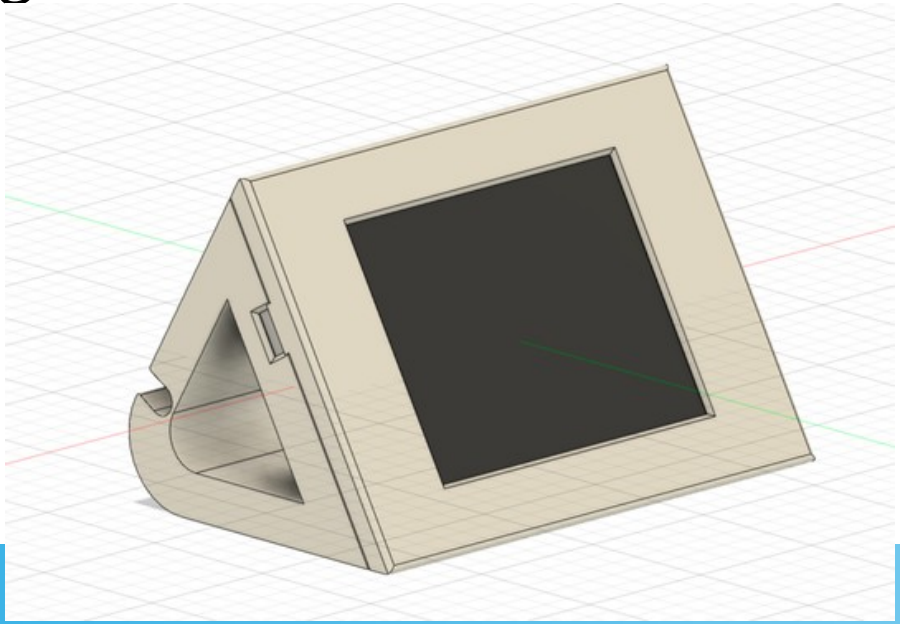
Convert competition device into a toy

Begin

- Device given during North Sec CTF competition in 2023 nsec.io

Competition ? NSEC ?

- What is Capture The Flag
  - Team game play during one weekend
  - Objective: Earn points by hacking software or hardware to retrieve a text Flag. *FLAG-THIS-IS-A-DEMO*
- What this device was used
  - Used to display information using a mesh network
  - Many flags available



The Board

- Device is `ESP32-2432S028R` as known as ESP Cheap Yellow Display
- CHIP `ESP32-D0WDQ6` with wifi and bluetooth.
- Development software ESP-IDF `Espressif IoT Development Framework`
- Constraint : Device was flashed using secure boot by NSEC team
  - They release source code with secure boot keys and building tools
  - Not reversible as when enabled it break a fuse to avoid stock back
  - Langage C++ and user interface with LVGL v7
  - Real Time OS with FreeRTOS

Investigation on repurpose item

- **FAILED:** Evaluated use for other stuff and cancelled as required to many research to use with secure boot as for:
  - ESPHome
  - ESP development
  - Secure Boot on ESP cannot be disabled

- **New objective:** Create a toy
  - Idea: Game score for mini figures
  - Planned of completion: 1 month
  - Real time: 4 months some hours on weekends

How it was

Take source code from NSEC team and modify it

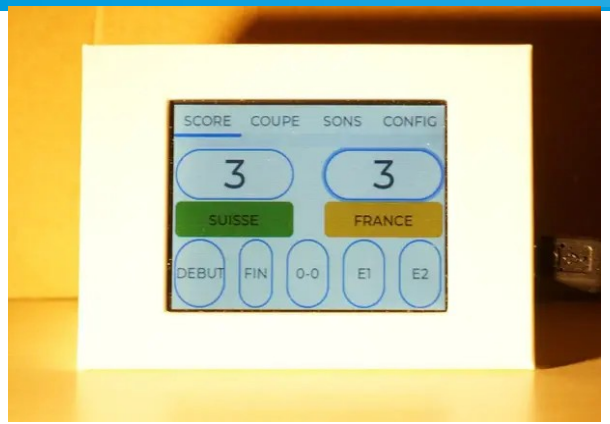
- Language C++ with ESP IDF
- Update scripts as not working as default
- Rebuild and flash to ensure everything works
- Remove components not used (Wifi, bluetooth, SD Card, ...)
- Build a new user interface
  - Modify existing one by try and fail
  - Use a simulator in C language
  - Adaptation for styling

What it contains / features*

- Scoring
- Colored items
- Teams selection
- Sounds
- Cups with different teams
- Use onboard led with color selection (already available)
- Screen backlight to preserve battery
- Store settings

DEMO

DEMO


SCORE
COUPE
SONS
CONFIG
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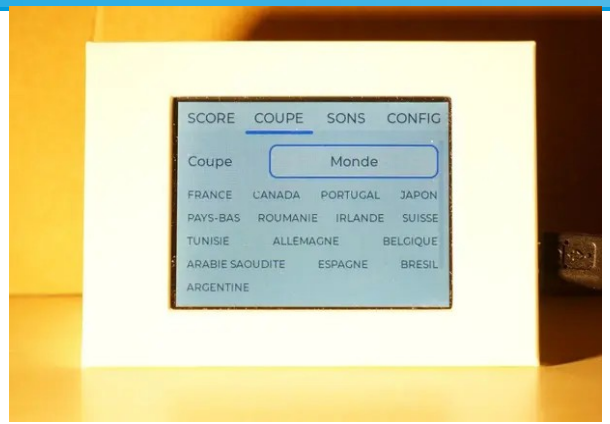
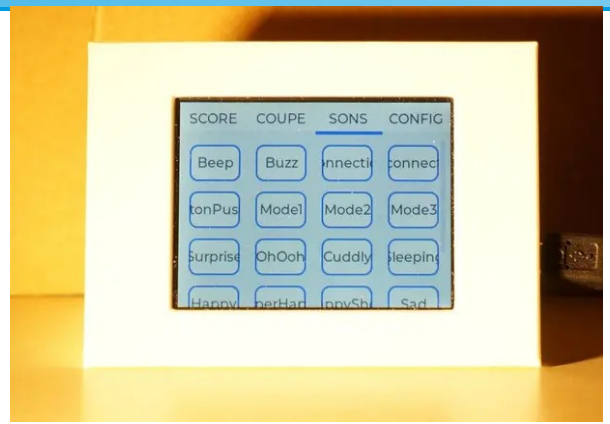
3
SUISSE
FRANCE
DEBUT
FIN
0-0
E1
E2


SCORE
COUPE
SONS
CONFIG
Coupe
Monde
FRANCE
CANADA
PORTUGAL
JAPON
PAYS-BAS
ROUMANIE
IRLANDE
SUISSE
TUNISIE
ALLEMAGNE
BELGIQUE
ARABIE SAOUDITE
ESPAGNE
BRESIL
ARGENTINE


SCORE
COUPE
SONS
CONFIG
Beep
Buzz
Mode1
Mode2
Mode3
OhOoh
Cuddly

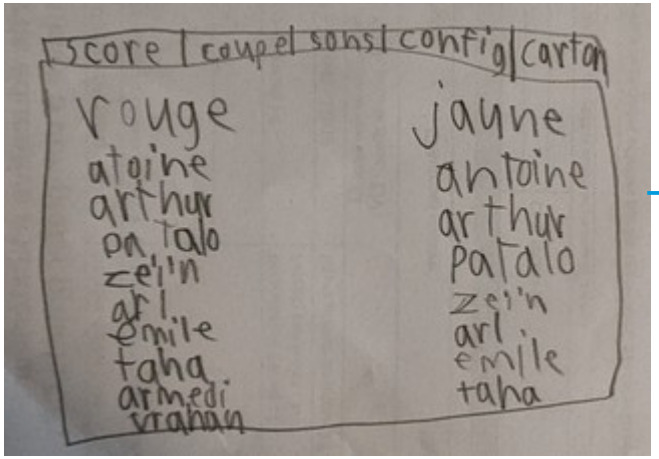
SCORE
COUPE
SONS
CONFIG
Couleur boitier

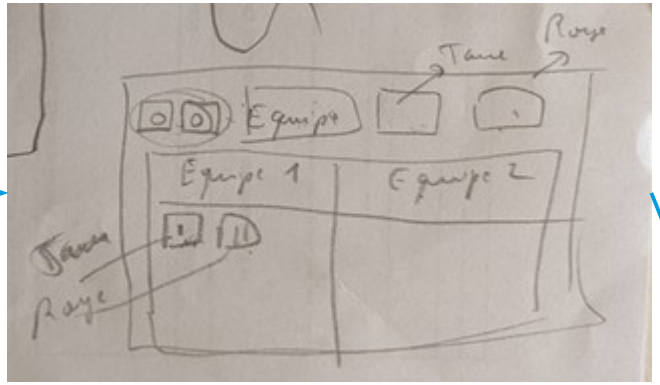
SCORE
COUPE
SONS
CONFIG
Solid
Blink
Breathe

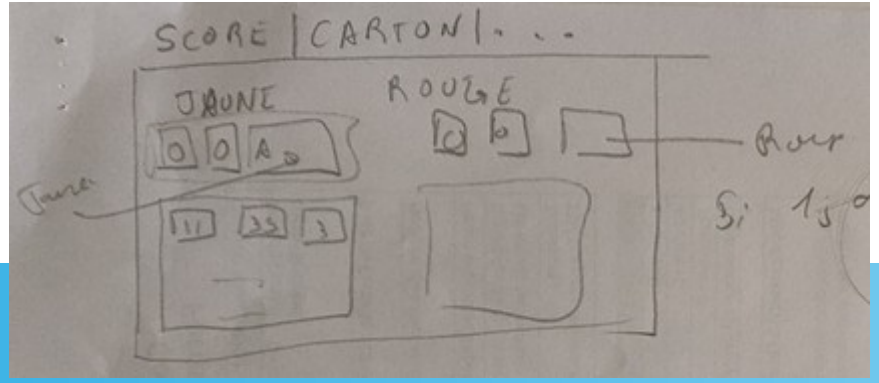
Last feature

On september for one month, addition of Yellow and Red Cards

Work with my kid on design



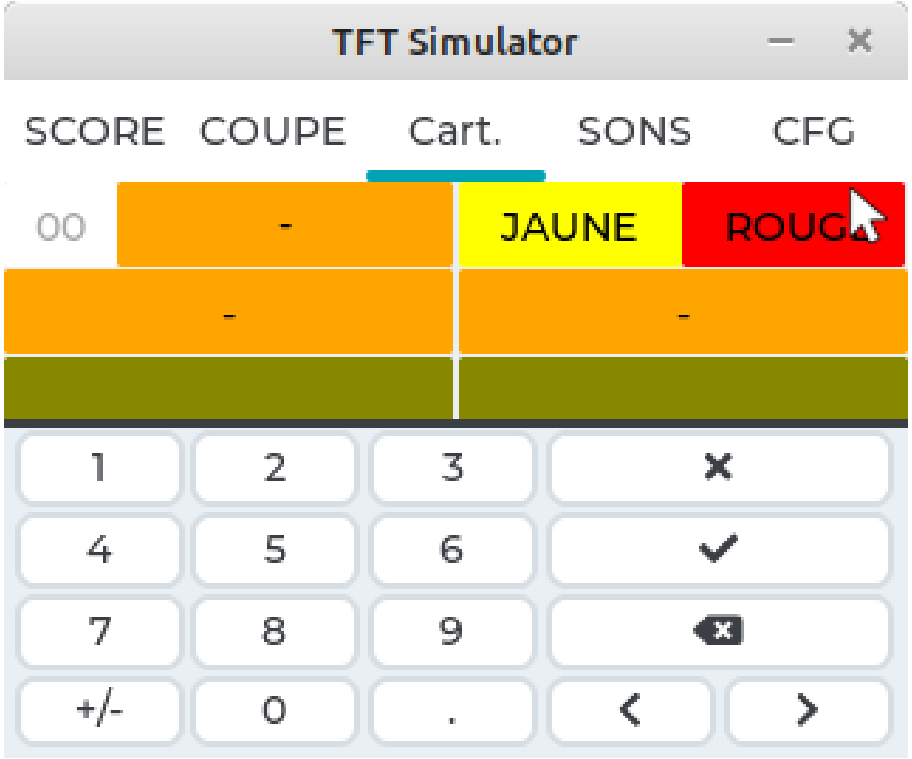




Challenge

- Use LVGL to display exactly at a screen position
- No borders to optimize screen space
- Remember the code structure

Samples


TFT Simulator
SCORE
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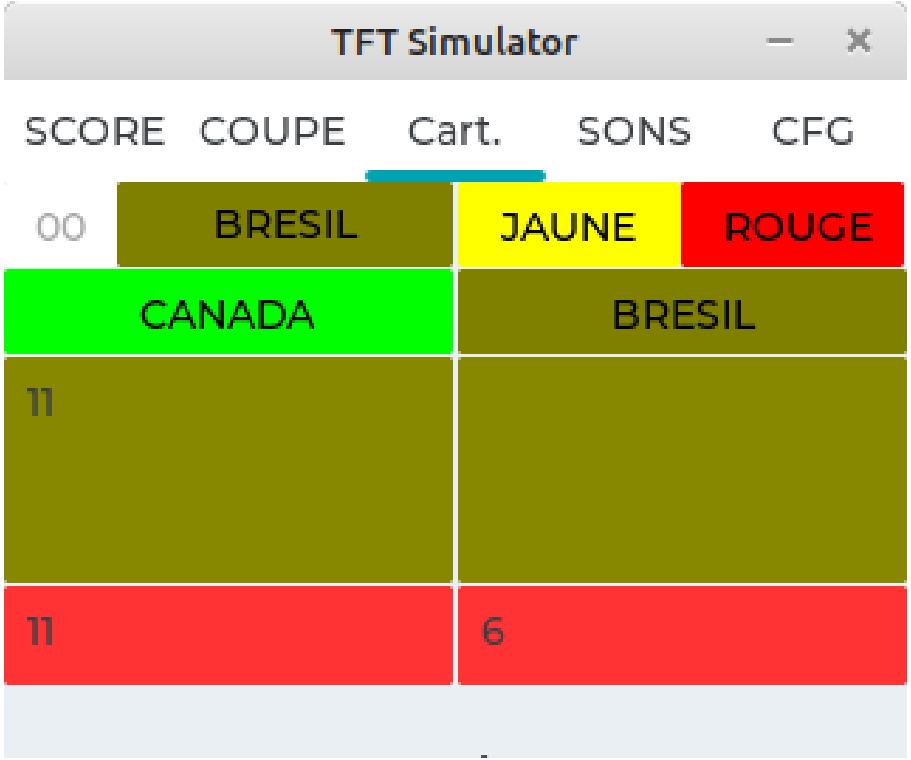
TFT Simulator
SCORE
COUPE
Cart.
SONS
CFG
00
BRESIL
JAUNE
ROUGE
CANADA
BRESIL
11
11
6

What I learned

- LVGL library for embedded device is simple and easy to use
- C++ is not as simple as I was aware
  - Random seeds
  - ESP Libraries modified by Espressif
  - Memories on C and C++ with #IFDEF to handle same code
  - ESP IDF make many under actions on compilation
- Final code is not optimal or even maintenable for a real product
  - **THIS IS A TOY !**
- Not able to fix: Remove some margins and borders

The End

GitHub: https://github.com/jingl3s/nsec-badge-display-other